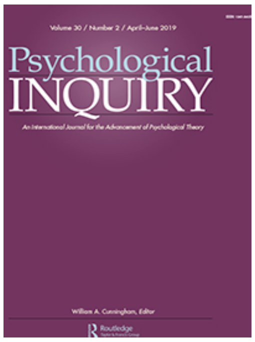

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TARGET ARTICLE

Archetypes Reconsidered as Emergent Outcomes of Cognitive Complexity and Evolved Motivational Systems

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ABSTRACT
Advances in contemporary cognitive science suggest that our internal representational systems are powerfully shaped by interacting evolutionary, developmental, and neuro-computational processes. Although Jung's archetypes of the collective unconscious are largely dismissed by modern psychological science, something very much like them emerges from the intersection of these perspectives. Functional analysis suggests that a variety of conserved systems—basic biological ones, like self-protection and mating, as well as more complex social ones, like cheater-detection—need to make use of more general representational systems (like face perception) to simulate and predict adaptive responses to recurring environmental problems. Furthermore, analogous to the capacity to develop language, these systems depend on specific input at critical developmental stages. Archetypes reflect the interaction of domain-specific challenges and domain-general simulations. They are dynamic patterns of perception, memory, and action, resonating with ancient motivational and emotional systems. They shed light on how the symbolic emerges from the subsymbolic. Archetypes are thus the natural consequence of our fundamental social goals playing out in three nested dynamics: the online representation of reality by mental simulation systems, the history of personal experiences that build a particular instantiation of these systems, and the evolutionary dynamics that selected the web of cognitive and affective capacities that all normally developing humans share. This modern elaboration of the idea of archetypes fuses disparate conceptual perspectives, provokes methodological reorientations, generates novel hypotheses, and will likely open whole new lines of integrative inquiry.

KEYWORDS
Archetype; Jung; evolutionary psychology; emotion; motivation; mental representation

Imagine sitting in a train as a menacing stranger boards. What thoughts and images come to mind? Now imagine, instead, a fellow passenger sneezing and sick, or one trying to flirt with your lover. Are the thoughts and images activated in response to these three strangers the same?

Individuals believed to afford different threats (or different opportunities) activate qualitatively different mental representations, and for good reason. Mental representations help guide actions, and because different threats and opportunities require different behavioral responses, it is important to differentiate in our representations between individuals (and "kinds" of individuals) who pose different threats and opportunities. Indeed, perceivers from very different cultural backgrounds are likely to employ fundamentally similar representations related to the basic problems posed by social life. Although a White U.S. university student and an Indian Hindu imagining a menacing commuter may recruit representations differing in certain surface characteristics (e.g., the dark skin of a racial minority member may come to mind for the American, whereas the headgear and facial hair of a religious outgroup may come to mind for the Indian Hindu), these representations will nonetheless be similar in many ways—in terms of certain demographic characteristics of the target (likely a young man), his facial expressions (scowling), his physical bearing (formidable), and so on. It was this recognition of great cross-cultural commonalities in the foundational contents of many social representations that led Carl Jung to propose a human tendency toward universal motifs of social representation—the archetypes.

Although long dismissed by cognitive scientists as new-age mysticism, Jung's archetypes warrant serious reconsideration[1] in light of emerging perspectives on mental representation (e.g., Freeman & Ambady, 2011; Pezzulo et al., 2013; Shepard, 1984) and evolutionary social cognition (Kenrick, Li, & Butner, 2003; Neuberg, Becker, & Kenrick, 2013). Although purely symbolic or propositional accounts of representation have considerable utility in certain domains, several factors suggest the need for a more dynamic and biologically grounded view of how we mentally represent our world and the complexities of social life. Across our evolutionary history humans have faced recurring problems of survival, sociality, and reproduction, yielding cognitive systems sculpted to effectively represent,

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[1]Although we have taken a different route to arrive at these ideas, particularly in the implementational aspects, the work of Anthony Stevens (1982) also considers the archetypes from an evolutionary perspective, and is well worth exploration, particularly for clinical matters.

simulate, and predict these fundamental challenges. At the same time, however, these systems also must be able to learn and incorporate culturally arbitrary distinctions (e.g., Becker et al. 2011; Johnson, Freeman, & Pauker, 2012) and be sensitive to variables embodied by social targets, such as their gaze direction (Adams et al., 2010), proximity to us, and our lines of escape from them (Cesario, Plaks, Hagiwara, Navarrete, & Higgins, 2010). Because purely symbolic or propositional accounts struggle to accommodate such requirements (as we review next), new models are needed. These representational systems must integrate (a) the deep structure of how we process the regularly occurring problems of (social) life, (b) the adaptive flexibility with which we represent proximate novelty and cultural innovation, and (c) developmental processes that weave these pieces of nature and nurture together. We suggest that representational systems designed to account well for these features (and others) will often bear fruit with more than a passing resemblance to Jung's original formulation of the archetypes.

To resuscitate and reinvigorate the idea of archetypes within cognitive science is therefore *not* to propose the typical new age interpretation—that we are all connected to a spirit realm of archetypal ideas by some kind of psychic umbilical cord. Soft appropriations of Jung's work in pop psychology and psychotherapy garner attention but stray from his original formulation. Rather, Jung was adamant that he conceived of archetypes as an inherently biological idea amenable to scientific investigation.

The idea of the archetypes emerges naturally from three increasingly influential perspectives in modern cognitive science. The first reflects applications of evolutionary biology to an understanding of the design of our cognitive processes and behavior. The second reflects evolutionary-developmental processes that guide learning. The third reflects new ideas about how mental representations are instantiated in the brain, not as symbols but as complex subsymbolic systems grounded in multimodal perceptual and motor activity. If we acknowledge that our thinking is often in the service of solving long-recurring social and biological problems, and attempt to characterize how the brain learns to simulate and predict the dynamic nature of these problems, something like the archetypes begins to coalesce. Even if Jung's precedent were absent, we suggest that there is now a need for something like the archetypes to characterize the deep structure of our social information processing.

We begin by reviewing Jung's original formulation, highlighting some central features neglected by recent treatments but which anticipate the present approach. We then further develop the evolutionary, developmental, and dynamically embodied features of a modernized account of the archetypes. We close by exploring broader implications of this approach.

What Are Archetypes?

The archetypes and the collective unconscious became a prominent part of Jung's thinking because of his psychotherapeutic work. Like Freud, Jung used dreams to look for signs of the causes of psychological disturbances. Unlike Freud, he saw a much richer set of motivations and recurring themes than just sex and aggression. The unconscious seemed to be filled with symbols of fundamental social roles (e.g., *Mother, Father, Child*) and processes (e.g., *Birth, Initiation, Marriage*). These "archetypes" appeared not just in dreams but also in the myths and folklore of all cultures. What could account for the near-universal prevalence of such ideas? Jung and von Franz (1964) postulated a biological and instinctive explanation:

> Just as the human body represents a whole museum of organs, each with a long evolutionary history behind it, so we should expect to find that the mind is organized in a similar way. It can no more be a product without history than is the body in which it exists. … I am referring to the biological, prehistoric, and unconscious development of the mind in archaic man, whose psyche was still closer to that of an animal. (p. 57)

Jung was not suggesting that that these "mental organs" are anatomically encapsulated or sculpted by natural selection alone but rather that they emerge in interaction with learning and culture. He also was not suggesting that specific, unitary images are biologically inherited.

> The term "archetype" is often misunderstood as meaning certain definite mythological images or motifs. But these are nothing more than conscious representations; it would be absurd to think that such variable representations could be inherited. The archetype is the tendency to form such representations of a motif—representations that can vary a great deal in their detail without losing their basic patterns. There are, for instance, many representations of the *hostile brethren*, but the motif itself remains the same. (Jung and von Franz, 1964, p. 58)

Thus, although certain specific contents of the images recruited by White American and Indian Hindu perceivers differ in ways that are culturally predictable, they nonetheless draw on evolved representational mechanics that evoke prototypes of the shared, universal archetype "hostile brethren."

Using more contemporary language, one can define archetypes as *core representations of social instincts:* Dynamic patterns of perception, memory, and action that develop ontogenetically (they can take culturally and individually variable forms) within phylogenetically sculpted channels that preserve a general species-typical form.

Three themes are central to our modern elaboration of the archetypes. First, archetypal representational systems have been selected to simulate and predict the basic problems of social life. They have arisen from coevolutionary dynamics stretching deep into our ancestral past, reflecting adaptations to the recurrent problems of hominid sociality and survival.

Second, archetypes depend critically on development and individual experience. They are systems that assemble themselves out of the buzzing and blooming experience of the infant; they adapt to changing goals across the life span; and they are prepared to take different paths in response to regularities of a particular ecology, such as resource-scarcity or sex-ratio.

Third, these archetypal representation systems are grounded in subsymbolic computational dynamics that are distributed across multiple modes of perception and action. This renders them exquisitely adaptive to the demands of an ever-changing environment, enabling them to draw flexibly on the stored experiences most relevant to the perceiver's present circumstances.

Precedents for each of these positions can be found in Jung's work (see Table 1), but even if he had never made such claims, they would nonetheless be well grounded by the intersection of evolutionary, developmental, and dynamical systems perspectives in contemporary cognitive science. We consider each of these components in turn.

Evolutionary: Archetypes Simulate and Predict Adaptive Responses to Recurring Social Problems

Archetypes arise from complex adaptive systems that have been selected to solve social and biological problems. Research and theory from ethology, sociobiology, behavioral ecology, and evolutionary psychology help characterize what these problems are and the nuanced behaviors to which they give rise.[2]

Very basic problems are dealt with by reflexes and fixed-action-patterns. As problems demand more flexible responses, though, basic drives that alert the systems to a need (e.g., glucostats detecting low blood sugar and giving rise to the feeling of hunger) need to use more general cognitive systems (e.g., learning, spatial maps) to achieve their goals (e.g., locate and consume food). Buck (1999) proposed that there is a hierarchy of such systems, organized by their increasing flexibility. Above drives he posited basic emotions, such as fear and rage, and higher still, social emotions like attachment and play. Indeed, with the growing cognitive machinery of our hominid ancestors, this top piece very likely initiated a new florescence of highly flexible social goals (e.g., Cosmides & Tooby, 2000). These goals possess some unique computational machinery but also make novel use of (a) domain-general resources like controlled attention and executive functioning, which are crucial for gathering goal-relevant information, and (b) older, more conserved emotional and motivational systems (e.g., fear, rage, curiosity; see Panksepp, 2004). Critically, these newer social systems involve representing other people, interactions among people, and scripts of interpersonal interactions that play out over time. These organizing systems serve *fundamental motivations* (Kenrick, Neuberg, Griskevicius, Becker, & Schaller, 2009)—the foundations upon which archetypal representational systems are constructed by evolution,

[2]There is already a strong precedent for such views in evolutionary biology, where it is axiomatic that higher animals develop perceptual mechanisms that discriminate the sex of conspecifics, differentiate among their ages, and infer their behavioral inclinations (Dawkins & Krebs, 1978; Dukas, 1998; Eibl-Eibeslfeld, 1975; Tinbergen, 1953). Many of these signal-detection feats seem to require some mediating representational system. For an antelope to identify conspecific breeding partners, and to recognize specific mates, requires an elaborate and flexible representational system, one that builds internal models and simulations to match, plan, and predict.

Table 1. Jung's writings anticipate modern evolutionary, developmental and subsymbolic approaches to cognitive science.

Evolutionary	Developmental	Subsymbolic
Evolved social instincts that represent, simulate and predict the fundamental goals of social life	Biologically prepared proclivities that are calibrated by experience and develop in stages	Inexhaustibly grounded in multimodally distributed, dynamic processing areas
"[The brain] has been built up in the course of millions of years and represents a history of which it is the result. Naturally it carries with it the traces of that history, exactly like the body, and if you grope down into the basic structure of the mind you naturally find traces of the archaic mind." (1976, p. 84)	"… not in the form of images filled with content, but at first only as forms without content, representing merely the possibility of a certain type of perception and action." (Jung, 1936a, p. 42)	Archetypes, in spite of their conservative nature, are not static but in a continuous dramatic flux. (Letters II, p. 165)
"… specifically formed motive forces which, long before there is any consciousness, and in spite of any degree of consciousness later on, pursue their inherent goals … the archetypes are the unconscious images of the instincts themselves, in other words, that they are patterns of instinctual behavior." (Jung, 1936a, p. 43)	"Primitive tribal lore is concerned with archetypes that have been modified in a special way. They are no longer contents of the unconscious, but have already been changed into conscious formulae taught according to tradition, generally in the form of esoteric teaching." (Jung, 1934, p. 5)	"… the archetypes of transformation … typical situations, places, ways and means, that symbolize the kind of transformation in question. Like the personalities, these archetypes are true and genuine symbols that cannot be exhaustively interpreted, either as signs or allegories." (Jung, 1934, p. 8)
"There are as many archetypes as there are typical situations in life. Endless repetition has engraved these experiences into our psychic constitution. … When a situation occurs which corresponds to a given archetype, that archetype becomes activated and a compulsiveness appears, which, like an instinctual drive, gains its way against all reason and will, or else produces a conflict of pathological dimensions." (Jung, 1936a, p.43)	"It is in my view a great mistake to suppose that the psyche of a new-born child is a tabula rasa in the sense that there is absolutely nothing in it. … It meets sensory coming from the outside not with any aptitudes but with specific ones. … These aptitudes can be shown to be inherited instincts and preformed patterns, the latter being the a priori and formal conditions of apperception that are based on instinct." (Jung, 1936b, p. 60)	"Not for a moment dare we succumb to the illusion that an archetype can be finally explained and disposed of. Even the best attempts at explanation are only more or less successful translations into another metaphorical language. (Indeed, language itself is only an image.) … And whatever explanation or interpretation does to it, we do to our own souls as well, with corresponding results for our own well-being." (Jung, 1941, p. 271)

Table 2. A sampling of fundamental motives, the (often overlapping) emotional systems they draw on, and examples of schemas and event-scripts that would reliably arise in the pursuit of system goals.

Fundamental Motivation	Examples of Emotional Systems Recruited	Example Archetypal Representations
Self-protection (Interpersonal aggression avoidance, but built on predator and injury avoidance)	Fear of injury; Anger if formidability of threat is low	Aggressive threat (likely male of the outgroup); Scripts for fighting, fleeing
Mating (acquisition, retention)	Lust, Fear of isolation; Downregulation of fear in mating contests	Mating ideals and rivals; Scripts for courtship and intrasexual contests
Parenting and nurturing	Attachment; Downregulation of fear in the face of offspring threat	Child; Scripts for caretaking
Status (pursuit, maintenance)	Rough-and-tumble play; Seeking (Panksepp's Dopaminergic reward circuit); Fear of status loss	Leader, Hierarchy; Scripts for supplication and usurpation
Social exchange	Fear of resource loss; Seeking; Anger in the face of cheating	Cheater, Valuation; Scripts for bartering

development, learning, and culture (see Table 2). The following are five fundamental motivations.

Self-protection (specifically, avoiding interpersonal aggression). All animals are motivated to avoid injury, and humans certainly share mechanisms that facilitate movement without falling, dodging pointy objects, even fending off animate predators. Because we also share a long history of aggressing against one another and engaging in organized group conflicts, our self-protective motivations also build cognitive systems to represent which humans are especially likely to pose aggressive threats; how and in which contexts to bluff; and when to preempt conflict, slink away when confronted, or engage in battle.

Mating. This is also a highly conserved motivational system. Although ancient instincts might provide the mechanics of mating, our dependence on culturally arbitrary courtship tropes requires a host of complex representational elements. We need to learn the mating and courtship systems of the specific culture we are born into—what is permitted and what is obliged. We need to assess the mate value of others as well as our own standing in the eyes of the mating pool. We must be attuned to the possibility of intrasexual competition for mates and be prepared to recruit behaviors (some perhaps from the self-protection repertoire, and some perhaps from the social affiliation repertoire) to compete successfully.

Parenting/Nurturing. Without nurturing, human infants do not survive, and they need far more sociocultural input than other animals. Thus, in addition to parenting instincts shared with less social animals, we possess a host of more elaborate social instincts designed to build the infant's social brain; these include "motherese," facial expression induction, and sensorimotor games. We also need to convey culturally arbitrary but socially crucial expectations. To do this we need to build internal models of the acts of nurturing and of the stages of development and build representations of healthy offspring and nurturing peers.

Status. We are born into social hierarchies. We must learn what constitutes status, how status is structured, how status covaries with the attributes and behaviors of others, our place in these hierarchies, and whether and how we might control changes in our placement within these hierarchies. We need to represent leaders and followers, hierarchical structure and fluidity, relative dominance and subordinate status, and supplicating and usurping behaviors.

Social exchange. We are interdependent creatures, needing to share resources and divide responsibilities to survive and thrive. We need to represent value and models of interpersonal exchange. We need to track reciprocity over time and permissible and obligatory actions.

These five social goals do not exhaust the list—notable others include disease avoidance, coalition building, and language-acquisition/exhibition—but they offer clear examples of motivations that require dynamic representations of persons and situations. They also reveal (at least) three central features.

First, social goals include subgoals and internal hierarchies, as revealed in the foraging and courtship models studied by ethologists (Alcock, 2009; Timberlake, 1993). For example, in animals that rely on visual cues, at different levels in the goal hierarchy, the same representational system allows the animal to perform the general task of rejecting a different species as a potential mate, the more specific task of assessing the mate quality of a conspecific, and the highly specific task of identifying one's own mate(s) among other potential mates.

Second, fundamental motivational systems are not neatly encapsulated but rather often interact with one another. They share resources and even influence one another. Panksepp (2004) reviewed animal evidence that, despite the fear and anger systems being functionally and anatomically differentiable, small amounts of fear activate the anger system, whereas large amounts of fear suppress it. This tipping point between fight and flight animates the human self-protection system as well, with the behavioral outcome depending on a context-specific calculus integrating information about a foe's formidability, status, and so on (Cesario et al., 2010). The ability of these systems to share information gathering and simulation resources provides more than just parsimony; it also allows for the emergence of new discriminations and central tendencies that may have little precedent in the ancestral past (e.g., via covariant learning; Stone & Van Orden, 1994).

Third, fundamental motivations may operate differently at specific developmental stages, leading to different phenotypes as a function of environmental input. Young rhesus monkeys reared in isolation are profoundly impaired in their social behavior as adults, but just 15 min of play per day with peers (and no interaction with the mother) largely eliminates these deficits (Harlow & Suomi, 1971). Likewise, in humans, evidence suggests that childhood adversity leads to a higher-than-average vigilance to social threats (Pérez-Edgar et al., 2010). In addition to certain phenotypes being fixed by early childhood experience, other social goals rise and fall with particular life stages: Consider the low threshold for intrasexual aggression that characterizes young, unmated men compared to the much higher threshold for older, mated men (Daly & Wilson, 1988).

These three features of fundamental motivations underscore one of the primary reasons we need a construct like the archetypes: The fundamental motives of social life are specialized to seek certain kinds of stimuli. However, postulating that self-protection is vigilant to "hostile brethren" begs the question about how such stimuli can be rapidly and accurately recognized. Most would agree with Jung that inheriting a fully developed image is absurd. How we develop and discover these releasers of our social goals thus becomes a crucial question.

There is another reason we need something like the archetypes: Empirical findings in social cognition increasingly reveal intersectionality effects that demand more flexible kinds of representational systems (and theorizing in general; Fiedler, 2004). Imagining an angry face will typically evoke a male face, and facial anger or masculinity facilitate each other's efficient recognition (Becker et al., 2007; see also Hess, Blairy, & Kleck, 1997; Marsh, Adams, & Kleck, 2005). Although socially transmitted stereotypes might be invoked as explanations of such effects, recent work shows that connectionist models with no such stereotypes can be trained on the same visual stimuli and produce the same kinds of outcomes (Zebrowitz, Fellous, Mignault, & Andreoletti, 2003). Similar intersectionality effects have been shown for race (e.g., Hess, Blairy, & Kleck, 2000; Hugenberg & Bodenhausen, 2004; Johnson, Freeman, & Pauker, 2012; Miller, Maner, & Becker, 2010), sexual orientation (Freeman, Johnson, Ambady, & Rule, 2010; Johnson, Gill, Reichman, & Tassinary, 2007), and age (Sacco & Hugenberg, 2009; Zebrowitz, 1996) using both static and dynamic face and body stimuli. Moreover, fundamental motivations can distort or sensitize perception and memory (e.g. Becker et al., 2011, 2014; Freeman & Ambady, 2011; for a review, see Neuberg & Schaller, 2014). Collectively, such findings indicate the need for representational systems like those we propose here, to parsimoniously explain these diverse interactions without also having to postulate an unreasonably large number of modules dedicated to producing such effects.

Fundamental motives are thus the foundations of the archetypes—constellations of special-purpose emotional and computational machinery designed to serve specific functions (akin to the subselves in Martindale, 1980, or Minsky's, 1986, "Society of Mind") but that critically require internal simulations and representations that are at least partially shared among motives/goals. Imagining "hostile brethren" brings online the "aggression avoidance" subself, which evokes both general face and body simulations, as well as more domain-specific emotional (fear, panic, perhaps anger) and cultural associations. Cultural innovations can of course select learning and training regimes that forge ever-new connections, continuously refashioning the outward form of the archetypes (see next). That said, the general tendency to form the archetypes is a part of our universal inheritance; it would be hard to imagine, for example, a future society in which the motif of interpersonal aggression has been completely purged.

Developmental: Archetypes Are Merely Capacities; They Depend Critically on Experience

The contents of archetypal representations reflect the fundamental challenges our ancestors faced, but they also must have the flexibility to vary with differences in life experiences. This is an often unappreciated aspect of evolutionary psychology—far from being opposed to learning, most theorists see it as critical to the development of mature cognitive systems (e.g., Cosmides & Tooby, 2000). Jung's early writings on the archetypes also reflect a weaving together of nature and nurture. Our modernization accordingly draws on and is quite consistent with the latest work in evolutionary-developmental thinking.

Consider Bjorklund's "evolved probabalisitc cognitive mechanisms." These constructs are developmental processes designed to solve recurrent ancestral problems that varied over time and across environments, such that one fixed phenotype could not reliably solve the problem (Bjorklund, 2016). Because development iteratively builds mechanisms and brain structures while receiving input from the environment (everything from nutrition to movement and information gathering), selection can act on regulatory processes sensitive to this input. Epigenetic factors can thus guide the same genotype to produce different phenotypes that reflect specific environmental challenges at a given time while leaving future generations other options when those challenges change (e.g., Frankenhuis, Panchanathan, & Barrett, 2013). Evolved probabilistic cognitive mechanisms thus aptly characterize the archetypes; they are capacities to represent but are filled with content (or even pruned away) by a given set of experiences.

The learning process itself is thus also critical to the formation and maintentance of archetypal representations. According to Csibra and Gergely's (2011) theory of natural pedagogy, adult knowledge of the social world is rooted in "powerful learning mechanisms that capitalize on innate biases, on statistical regularities extracted from the environment and perhaps even on capacities to construct new representational systems" (p. 152). The innate biases include a bias to watch others, to follow their gaze, to assume that their vocalizations are related to their gazing and pointing, that these vocalizations refer to "kinds" rather than unique instances—in short, for infants to assume they are being taught. A handful of such biases can go a long way toward assembling the mature categories that most of us possess.

Perspectives such as these entail that the archetypes cannot be reduced to either nature or nurture alone but arise from cycles of their intermingling. Analogous to the development of language skill (e.g., as characterized by Newport, 1990), archetypal representational systems likely assemble themselves in stages, adding new culture- and experience-specific knowledge on base representations and earlier developments. Key to this development is the tandem emergence of a nested set of general representational systems (face, body, dyadic interaction), as well as a differentiation of social goals that appears far more domain-specific. Table 3 presents some of the general representational workspaces the developing child is likely to assemble, along with the goal systems that emerge and the specific kinds of representations they require.

Although it is beyond our current scope to fully flesh out the development of archetypal representational systems, we provide a preliminary description to highlight key issues to be considered.

Infancy. Representing faces is one of the key tasks of the infant. The face-representation system may involve an initial visual template to identify facelike shapes along with an attentional proclivity to study such objects (Johnson, Dziurawiec, Ellis, & Morton, 1991), but environmental input is critical to carve out the categories (e.g., female and male, angry and happy) that will later play such an important role in person perception. Internal models of face representation may also depend on certain hardwired proclivities, like using instinctive templates in the olfactory systems to distinguish women from men (or mother from all else) and kin from non-kin. The fundamental motivations of adult life are still in a primordial state. However, differentiable emotional systems, especially those identified by Panksepp (2004), are already engaging somewhat discrete operations, providing an underlying texture to the overly simple general categories of "positive-negative" valence. The development of emotional categories is likely also facilitated by the infant's ability to perceive and mimic facial displays of emotion (e.g., Field, Woodson, Greenberg, & Cohen, 1982); facial mimicry is likely critical to the catalyzing of mirror-neuron systems.

In infancy, then, the earliest conditionings of archetypal representational systems begin as very simple, face-based social discriminations. Especially prominent in infant development should be the emergence of an archetype of caretaking (Jung called this the *Mother*) centered on the face, as well as basic sensorimotor goal-approach–avoidance action scripts (e.g., the schemas discussed in Lakoff, 1988). Given that alloparental care is a regularity across most human societies (Hrdy, 2009), the general face representation system very likely instinctively seeks the differentiation of caretakers, which provides experiential foundations for archetypes of male and female. Covariation with features of emotional arousal will color these figures, and cross-cultural regularities begin to provide these archetypes some of the consistency that later will be taken as evidence for universality. Seeds sowed at this stage will also influence goals that are only latent, like mate-quality and dominance appraisal systems.

Table 3. As the mind develops, it requires increasingly complex general representational systems that can best serve the adaptive goals of the particular developmental stage, and the discriminations upon which the satisfaction of those goals depend.

Developmental Stage	General Representational Systems	Goal Systems	Discriminations	Example Archetypal Representations
Infant	Face	Seek nurturance, avoid harm	Male and female; emotions	Caretaker
Toddler	Body	Aggression and disease avoidance	Otherness (outgroups); Bodily Skills	Stranger
Linguistic child	Theory of mind	Skill and culture acquisition, imitative play	Young and old; expertise; personalities	Peer and wise elder; apprenticeship
Social child	Multiperson interactions	Social exchange, status	Friend and enemy; social hierarchy	Cheater/trickster; exhange
Pubescent	Tribe and society; pool of allies, mates and rivals	Mating and parenthood	Attractiveness; availability and Interest	Mate/Rival; courtship; birth

Note. The child's growing abilities to interact with and act on the world require increasingly complex domain-general representational capacities and discriminations. Example archetypes are listed, but we emphasize that these central tendencies are merely snapshots of a much more dynamic—Jung called it "inexhaustible"—substrate.

The Toddler. As sensorimotor skills enable walking, representing bodies and their movement becomes essential. Just as faces have male and female prototypes, so do the shapes and movements of bodies (Kozlowski & Cutting, 1977; see also Johnson et al., 2007).

As the dimensions of person construal begin to differentiate themselves, the stranger anxiety (Ainsworth, Blehar, Waters, & Wall, 1978) experienced by most children sets the stage for representations of other social groups, and the archetype of the "other" becomes linked to social threat. Cultural differences in childhood attachment styles (van Ijzendoorn & Kroonenberg, 1988), however, suggest that the environment is already playing a key role in tuning these archetypal representational systems. We see here the first differentiations of simple harm-avoidance motivations as newer constellations of representation and action develop differentiated foci on more specific goals like avoiding falls; avoiding aggression; and, perhaps this early, avoiding disease (e.g., food aversions). Specific kinds of childhood adversity may also reliably evoke specific developmental trajectories; for example, physical abuse might tune the general face and body representational systems to be vigilant for signs of anger (e.g., Belsky et al., 2009). These systems may also be scaffolded on other systems: Aggression avoidance could be a modification of injury and predator avoidance in the self-protection domain, or instead converge more on protection seeking, a differentiation of the infant's nurturance-seeking motivations.

Of course, there are many dimensions and degrees of otherness, and cultural markers must be learned that provide new types of classifications. For example, social learning and imitating the parents likely lays the groundwork that will later differentiate among the types of threats or opportunities that other groups pose. A vigilance to signs of aggressive threat likely emerge first, but the groundwork for detecting cues of preferred trading partners, nurturing others, or particular types of skill must also be developing. A common space where bottom-up information mingles with different social goal systems is thus critical. Another feature of these general representational systems as they struggle with new motivations coming online is interpolation—generating representations beyond the boundaries of the child's day-to-day experience. Such interpolation may explain the development of fairy-tale imagery (see Jung, 1948) and anthropomorphisms in the child's attempt to understand the unknown. If aggressive threat detection, for example, shares something with systems attuned to predatory animal threats, such considerations may lay the groundwork for the tendency to make beastly and supernatural ascriptions to the outgoup.

The linguistic child. Language hints at an inner simulation and classification facility that tracks an increasingly complex social world. It also, reciprocally, suggests that others have a mental life, and a new general representational space to simulate other minds needs to develop. This is considerably more abstract than the representations of faces and bodies, but it is likely anchored in these earlier systems, as well as in the child's developing sense of his or her own interiority and its opacity to others. The child's theory of mind begins to coalesce, affording predictions about the inner lives of others along with their unique perspectives, especially taking into account the knowledge that might be hidden from them. Language development and household technologies and crafts also bring forth appraisals of skill and specialization. The representations depend critically on actions; consider that the actions a child models after adults precede the language to describe these actions by about 12 months (Glenberg & Kaschak, 2002). At this stage, we should see status distinctions and skill specializations begin to differentiate the ingroup, and these differentiations may lay the groundwork for a distinction between peers and elders. Age likely forms the basis of early comprehension of status differences as well, setting the stage for the representation of social hierarchies and social exchange. Play is an incredibly important motivational system (Panksepp, 2004) that seeks teachers as well as peers and takes forms that prepare the child to develop the skills and representational systems of adults.

The social child. Once a child can represent, simulate, and predict faces, bodies, and minds, it is possible to represent social interactions. The dynamic nature of the archetypal representational systems is now essential; it is not just imagery but event-sequences and scripts that need a general representational space, scaffolded upon and coordinated with the other systems that have already developed. Systems designed to avoid new kinds of dangers (e.g., resource loss) and pursue new opportunities (e.g., status) now come on line. Only at this point does it make sense to reason about social obligations and permissible actions, to look for cheaters, and to diagnose clues about the motives of others (Cosmides, 1989). Detecting liars and cheats becomes of real interest at this age (e.g., Harris, Nunez, & Brett, 2001) and appears to correspond to dedicated neural systems. Neuroimaging research now suggests that different areas in the prefrontal cortex appear to underlie the functionally distinct components of reasoning about social exchange. The ventrolateral prefrontal cortex (vlPFC) has been shown to be active when reasoning about violations of social contracts (including prohibitions as well as obligations; Fiddick, Spampinato, & Grafman, 2005), whereas the dorsolateral (dl) PFC instead appears to be more engaged by questions of what is permissible within these obligations. It is intriguing that the vlPFC is developmentally innervated before the dlPFC. From the standpoint of functional specialization, it makes sense that we first develop the ability to think about what society requires of us and only then develop the ability to simulate and predict the consequences of bending those rules.

Rough-and-tumble play begins to take on forms that practice adult roles and establish physical dominance hierarchies. Panksepp (2004) used comparative neuroscience to show that a somewhat separate aggression system underlies intrasexual competition. It has many of the same overt features as anger but is released by different environmental triggers and does not seek the same kinds of resolution (e.g., It more readily subsides when the "foe" makes a gesture of submission). These conflicts can be overt, like playful wrestling, but they also involve covert and deceptive conflicts that involve a maturing representation of others' mental states. The result of these playful conflicts is the ability to simulate and predict the outcomes of such conflicts and an embodied understanding of status. Archetypes of dominance, leadership, and rivalry are thus likely to arise.

Pubescence and adulthood. Puberty brings with it the emergence of representational systems dedicated to finding and attracting the best possible mates. Rough-and-tumble play has already set the stage for intrasexual mate competition, especially for boys, and children learn their relative standing on a wide variety of social dimensions, including those that will set the stage for self-assessed mate value. The ability to enter the reproductive arena also influences status striving, coalition building, and skill display, all of which had roots in childhood but take on new significance once sexual reproduction is possible. Universal aspects of mate evaluation, like waist–hip ratio or facial-feature configurations, pose a particularly interesting topic of study for evolutionary cognitive science. How does a cascade of hormonal changes at puberty transfigure the way that attention now focuses on information that distinguishes not only the two genders but also their fertility and mate quality? Jung noted that "the whole nature of man presupposes woman, both physically and spiritually. His system is tuned into woman from the start, just as it is prepared for a quite definite world where there is water, light, air, salt, carbohydrates, etc." (Jung, 1940 p. 188). A woman's mating system has been similarly prepared, and LGBT variations will also draw on many of these same preparations, albeit in different combinations. The archetype of the ideal mate emerges, but several important questions remain unexplored. Most critically, how much of this is based on our concrete experiences with the desired sex, and how much is based on evaluation parameters that have been set by natural selection? Phenotypic differentiation also occurs at a biological level: Girls raised without a father show earlier onset of puberty and mating ideals that prioritize "good genes" over stable resource provisioning (Ellis, 2004). This underscores the fact that archetypal representational systems, like the language learning faculty, have expectations about the social information they should encounter—in this case, for a father figure as well as a mother—and that failure to find this information in the environment recalibrates the system at a fundamental level, speeding up maturation.

Although most representational systems and domain-general processing systems are relatively well-developed by the time a person can reproduce, parenting does bring with it new social goals that make new demands on the representational systems. The birth of a child may represent a qualitative shift in how representational systems are used and organized, or it could be a fairly continuous development of the care system that is evident in young children. Regardless, the system's more basic purposes are to represent perceptions and actions that encompass nurturance and child-rearing. The child is an integral part of simulations that not only facilitate childcare and welfare but teaching and training more generally. It is intriguing that this system can influence calculations about social cheating: Being baby-faced can generate more trusting evaluations and behaviors in interaction partners (Zebrowitz, 1996).

The archetypes of the adult thus have a very complex etiology, but they nevertheless generally converge on a common set of forms that reflect the regularity with which our ancestors faced a common set of challenges. Table 4 lists some of the outcomes of the representational systems that correspond to the fundamental motives we listed at the outset.

Plasticity and Culture. Two points about the development of such systems should be highlighted. First, there are sometimes facultative mechanisms that arise only in response to specific challenges. We may see reliable bifurcation points in development—situations in which an environmental variable, like safety, can lead to two phenotypes—one adapted to harsh environments and one better suited to nurturing environments. Belsky (e.g., Belsky et al., 2009) has postulated that such plasticity is often seen in the Gene × Environment interactions underlying mental disorders. For example, individuals carrying a particular serotonin transporter gene show higher levels of anxiety and attend more to social threats, but only when they experienced adversity as a child. The identical allele leads to greater resiliency when raised in a nurturing and socially supportive environment. Archetypes might show a similar endophenotypical bimodality across persons, despite their sharing the same genotype. In a person from an adverse environment, the central tendency of "leadership" may have aggressive qualities; for a person with a very secure upbringing, "leadership" may instead take a gentle and reflective form.

Second, if recurrent social problems and goals have sculpted the representational predispositions in the brain, it has done so in the context of very different cultures, a process of dynamic optimization (Frankenhuis et al., 2013). As previously noted, a key aspect of the plasticity of the archetypes is the way they preserve functional consequences in light of cultural variance. Systems of organizing and representing hierarchies, outgroups, sex roles, and many other factors may find some explanation in ecological commonalities as well. Commonalities in representations will sometimes reflect dominant technologies needed to respond to local ecological challenges and opportunities. Joseph Campbell's (1988) *Historical Atlas of World Mythology*, for example, sought to discriminate common themes in hunting-based versus horticultural societies. Cross-cultural work is thus essential. It allows us not only to identify universally desired characteristics but also to get beyond our own media-saturated environment, which is unlikely to fairly weight the role that experience with others played in small-scale societies. Much remains to be done to integrate the diversity of human ecologies into the present framework.

Dynamically Embodied: Archetypal Representational Systems Are Dynamic, Multimodal, and Subsymbolically Grounded

General learning and abstraction algorithms can readily generate culturally arbitrary concepts and prototypes—for

Table 4. A broader set of Archetypes that stem from the present fusion of domain-general representational capacities and domain-specific motivational systems.

Fundamental Motivation	Vigilant for	Archetypal Representations
Self-Protection	Aggressive threats (young males, outgroup); Anger; Coalitional Allies (strength, bravery); Safety and Cover. All calibrated by one's own formidibility.	Enemy, Ally, Warrior, Hero, Coward, Fighting, Fleeing.Relations to predation and injury avoidance, mating related intrasexual-aggression.
Mating	Potential Mates; Attractiveness; Rivals; Intrasexual Conflict; Flirtation and Interest; Infidelity. All calibrated by one's own mate value.	Male and Female, Beauty, Courtship, Mating Rival, Mating Contest, Marriage, Faithfulness. Distinctions among acquisition and retention; deep roots in animal behavior.
Parenting and Nurturing	Neotony; Signs of Helplessness, Illness. All calibrated by kinship.	Child, Caretaker (Mother, Father), Birth, Soothing, Protection. Readily coopts and overrides other systems, e.g. protection of others often trumps self-protection.
Status	Biological markers of dominance; cultural indicators of prestige; Weakness in leaders; support among allies. Calibrated by one's own position in the hierarchy.	Leader, Follower, Hierarchy, Usurption. Moderates many other vigilancies, e.g. who is an appropriate mate, to whom aggression can be displayed.
Social Exchange	Rule violation (prohibitions and obligations)., deals, rarity. Calibrated by one's need, social position.	Thief, Bartering, History of Exchange. Abstractions of value and social obligation; may be related to exogamy.

Such systems are vigilant for goal-relevant information (the presence of which can act as an arouser of the motivational system in question), but are also calibrated by situational and contextual factors, which sometimes leads one Archetype to override another.

utensils, hairstyles, and anything else evolutionarily novel that needs to be categorized. Although the archetypes are not novel—they have been biologically prepared by motivational and emotional systems, and reliably develop, even when the environment is impoverished—the way their representations are implemented shares qualities with more arbitrary representations. In fact, their openness to cultural forces mandates a flexibility that underscores shortcomings with traditional models of symbolic representation (despite readily giving rise to symbols). Archetypal representational systems must promote not only static imagery and symbolism but also dynamic goal-pursuit, predictions of outcomes, and simulations of counterfactuals. We thus suggest that the archetypes of social life are likely implemented via subsymbolic, multimodal, and grounded representations.

The precedent: Categorization, prototypes, and exemplars. Questions about how we categorize and represent the world have been explored at least since Plato; his theory of a world of ideal forms is still an influential attempt to understand the foundation of generic concepts and categories. Kant's notion of "schemata"—representing those features that all members of a class share—is another. In fact, different readings of Kantian schemata have provoked critical developments in cognitive science, such as Bartlett's (1932) work on memory. Problems with providing necessary and sufficient definitions of schemata/categories, however, also led thinkers like Wittgenstein to question the possibility of ever exhaustively defining the features of basic concepts.

In cognitive psychology, such thinking gave rise to the "prototype" as a way to capture the central tendency of concepts (Rosch, 1973, 1978). If most categories cannot be defined by boundary conditions (i.e., definitions), or if the boundaries are fuzzy (e.g., between "boy" and "man"), then equating them with the central tendency of the category may better depict the concept being represented. Prototypes are the most typical members of a category (and may sometimes even correspond to members never encountered in reality). Prototype-based explanations of categories have fuzzy boundaries and graded memberships. For example, for most of us, a sparrow is a more central example of "bird" than is an ostrich. This has measurable consequences for our thinking: For example, when discriminating birds from other animals, people are faster and more accurate with sparrows than ostriches (e.g., Rosch, Mervis, Gray, Johnson, & Boyes-Braem, 1976). By the same token, our prototype of aggressive person will more likely be a large male stranger than a short grandmother.

It would be a mistake to see archetypes as merely prototypes, however. Prototype models can be invoked to account for all sorts of arbitrary categories, learned in one culturally contingent environment but not another. Archetypes are instead universal tendencies to seek and build functional correlates of recurrent social motivations. They seek to arise even in the absence of environmental input, though perhaps in a stunted, distorted form.[3] Whereas most readers can

[3]Consider a comment by Jung on the archetype of the caregiver, which he calls the Mother: "The more remote and unreal the personal mother is, the more deeply will the son's yearning for her clutch at his soul, awakening that

envision a prototypical smartphone, it is clearly a modern development with few historical antecedents. Archetypes, in contrast, reflect ancient and recurring social motivations. Even if they are filled with novel technological experiences, they reflect ancient urges; Facebook, for example, might now influence our archetype of friendship, but equally its success is predicated on long-standing drives to seek (and internally represent) affiliations with others. Archetypal representations thus go beyond the prototype in that, unlike highly culturally specific prototypes, the archetype of aggressive threat is something that reliably asserts itself (even in safe ecologies). It searches for experiences that can build representations, which will serve as triggers/releasers for the motivational system, and influences other seemingly unrelated acts of social judgement and inference (as well as dreams and mind-wandering).

Equating categories with central tendencies, however, loses potentially important information about the variability in the category instances. If our representational systems instead preserved experiences of the actual exemplars, each exemplar could leave behind traces that allow much more flexibility in extracting central tendencies subject to other (multiple) constraints. Indeed, many empirical findings suggest that fairly detailed traces of exemplars are stored in our representational systems. For example, people can take account of correlations among features of the exemplars when categorizing (e.g., Medin, Altom, Edelson, & Freko, 1982). Moreover, category boundaries and prototypes can shift if we are asked to take different perspectives (e.g., Barsalou, 1988). An American thinking about food might reveal that "bread" is prototypical, but, if asked to take the perspective of someone in Southeast Asia, can readily perceive the centrality of "rice." Such results are difficult for prototype models to explain without vastly increasing the number of prototypes that must be available to us in a latent form, many of which may never be needed in practice. Traces of exemplars, on the other hand, allow sampling of our actual experiences, and this allows more flexibility in taking different perspectives based on those experiences. Archetypal representational systems also need this flexibility. What kind of representational systems can attain this flexibility while still having the central tendencies that allow generalization and response gradients as prototypes do?

Subsymbolic Representation. Connectionism and parallel distributed processing models introduced a different approach to mental representation (Bechtel & Abramson, 2002; McClelland & Rumelhart, 1986; Smolensky, 1991). From this perspective, imagery, symbols, and simulations are patterns of activation distributed across neurons (or processing units), yielding representational systems that can both preserve episodic details of our unique experience while allowing the extraction of central tendencies and prototypes when required (Knapp & Anderson, 1984). If humans share common biological and social goals that influence attention and information pickup in consistent ways, then even these highly empiricist models will generally converge on forms in the adult mind that yield cross-culturally consistent archetypes. These architectures also enable highly adaptive representations of evolutionarily and culturally novel experiences. Some advantages of this subsymbolic approach to mental representation follow.

Content-addressable memory. Traditional models of memory are like libraries, where each piece of information has an addressable location. Subsymbolic systems instead have a distributed structure in which any feature can be queried to find other exemplars that have that feature (e.g., McClelland & Rumelhart, 1981, 1986). Conjunctions of features often allow us to narrow even further to the precise exemplar desired, and this is accomplished far more rapidly than serially searching a library. This affords another valuable property: In "noisy" environments, patterns can be completed with only partial information. This can lead to biased performance—and indeed, many biases in social cognition can be elegantly explained with connectionist systems (e.g., Freeman & Ambady, 2011)—but the benefits of rapid categorization also accrue (e.g., Fiedler, Kutzner, & Vogel, 2013).

Context-sensitivity. Because contexts matter, our mental representations need to be sensitive to them. A young man walking toward you down a dark alley at 2:00 a.m. means something different—and requires a different suite of perceptual, cognitive, emotional, and behavioral responses—than does the same young man walking toward you on Main Street at noon. Seeking evidence that one's attraction to another is requited requires different information in different contexts—a longer-than-normal glance at a formal party, a racy eyebrow flash in a darkened cabaret. Mental representations facilitating our ability to draw appropriate inferences need to be sensitive to contexts. Subsymbolic systems naturally yield this dynamic context sensitivity.

Soft constraints. Context dependency reveals another advantage of such systems. The conditional rules of logic reflect hard constraints—if P then Q, no exceptions. Because connectionist systems naturally incorporate context-dependent behavior, they can give rise to softer-graded responses while still giving rise to rulelike behavior in less ambiguous circumstances. If another person is seen caressing my mate, this might trigger an emotion like jealousy, which in turn might evoke a host of other responses including potentially costly conflict with the rival (or mate). But if the caresser is a child, jealous and aggressive responses are heavily downregulated (Sagarin, Becker, Guadagno, Nicastle, & Millevoi, 2003). In a symbolic system, such rule exceptions may be so numerous as to make learning impossible. Connectionist systems can easily model complex rule and exception sets and can learn these from manageable sets of experience. This has been robustly demonstrated for language learning (Plaut, McClelland, Seidenberg, & Patterson, 1996), in which grammatical rules emerge as well as predictable types of error commonly observed in young language learners (e.g., past-tense regularization).

primordial and eternal image of the mother for whose sake everything that embraces, protects, nourishes, and helps assumes maternal form, from the Alma Mater of the university to the personification of cities, countries, sciences and ideals."

Consider how a "moral" grammar emerges (Mikahil, 2007). We understand violations of social rules by internally modeling relations between agents and patients, which resonate with archetypal representational systems. But neither these rules nor the archetypes that animate them involve hard or discrete constraints. Furthermore, the interaction of agents and patients changes over time, such that what starts as innocuous and permissible can cross into a range in which we seek to explain our jealous reaction by appealing to prohibitions. Such dynamics could be accounted for by complex rule sets that have in some way been hardwired, but a subsymbolic representational system tuned by experience entails content addressability, context sensitivity, and soft-constraint satisfaction—a far more parsimonious explanation than a combinatorial massive set of rules.

Learning in subsymbolic systems. Many of the simpler connectionist models can readily explain how we learn categories in a supervised fashion, where a teacher reinforces the correct decision. Recurrent networks (which include bidirectional feedback and feedforward connections) are additionally capable of unsupervised learning and category induction and can do this with time-varying streams of information. Learning in such networks has several interesting implications for archetypal content. Early stages of experience (training) can lead to a network producing one diffuse central tendency of a category, whereas with additional experience more nuanced categorical discriminations can be accommodated. In the case of representing "hostile brethren," for example, the earliest forms may be very general—fairy-tale figures and bogey men—but as the child develops, she incorporates elements from culturally defined outgroups and strangers, and then later will come to possess different content-addressable forms when aggressive versus economic versus disease threats are being considered. Piaget's distinction between assimilation and accommodation finds a natural home in the operation of these models (Karmiloff-Smith, 1996).

Adaptive learning in such models depends on attention, which we postulate is biologically prepared to build the archetypes of social life. Modelers often build attentional parameters into recurrent networks to generate more complex organization of representations, appropriate to dynamic flows of information like those encountered in speech perception. For example, in sentence processing, recurrent networks can build connections based on local covariance early on (letters next to one another), then expand the "attentional window" to capture covariations across broader swathes (words, sentences) of space and time (Elman, 1990). These later stages of the model's training and development encode higher order covariations among the local covariations discovered in the earlier developmental stages of the model, which now act as units. Csibra and Gergely's (2011) model of learned pedagogy similarly privileges attention, maintaining that children are designed to attend to the words and phrases that adults use with them as defining the most useful conceptual distinctions in the blooming, buzzing confusion that surrounds them, moving from basic level to more general and specific concepts as they age.

Consider how such models might be extended to our learning of flirtation and courtship rituals. The behavioral "atoms" like eyebrow flashes and hair flips (Moore, 1985) are learned early on, before puberty, and play multifunctional roles. It is only later, however, when they are woven together in particular sequences and environments, that they take on the signals of courtship. Recurrent connectionist models of speech perception thus provide key insights into how archetypal representational dynamics might develop, diverge, and transform themselves over time.

Grounded representations. Although they are not necessarily connectionist, "situated" (Greeno, 1994; Smith & Semin, 2007) and "embodied" (Barsalou, 2009; Glenberg, 1997) perspectives on cognition are inherently subsymbolic and characterize the way in which representations can be grounded in experience. These perspectives emphasize representations and simulations that arise from the coactivation of perceptual and motor neurons distributed across multiple modalities (i.e., spanning the senses but also including interoceptive, emotional, and motivational states). Shepard's seminal work on mental rotation is a good example of how internal representations can be grounded in action (reviewed in Shepard, 1984). The activity patterns have their beginning in immediate experience and simple perception-action relationships, but higher order conceptual simulations can also arise. Barsalou (2009) suggested that the idea of a bicycle, for example, amounts to reenactments of our past perceptions of bicycles as well as motor actions of riding, hefting, repairing, and even crashing on them. Memories are stored, at least in their most concrete parts, where the perceptual and motor processing was originally done. Barsalou's "bicycle" simulator system is the aggregate of these multimodal patterns and proclivities, and it has many interesting properties that both allow us to act on the world and enable us to query for central tendencies to capture prototypical "bicycles." Of particular interest is the context sensitivity that grounded representations allow: We can simulate mountain bikes versus road bikes, children's bikes or broken bikes, accidents or victories, all by allowing the simulator to be influenced by context and/or activation in other perceptual, interoceptive, and motor systems.

These perspectives can be profitably extended to the social domain. The subsymbolic elements in the "face" simulation system will have a history of covariation in multimodal activity, which can include the activation of fundamental motivational/emotional systems. Activate a sense of self-protection and a very different face will likely be simulated than if you activate a mating motive. Activate the social exchange system and we may become vigilant to detect rule violators but add a high-status person to the situation and rule-violation detection may well be suppressed (e.g., Cummins, 1999). If archetypes arise from grounded simulations innervated by fundamental motives, they inherit these many benefits.

Grounded simulations also afford prediction (Barsalou, 1999, 2005). Simulations over time allow us to predict how to pick a bicycle up off the ground, adjusting our effort as we get feedback from the muscles and the visual system that tells us the wheel is stuck on something. Grounded simulations allow us to plan for the future ride home and enable us to rapidly recognize unexpected obstacles in the path and react appropriately (Glenberg, 1997). Similarly, the grounded script of courtship likely allows young lovers to adjust quickly to the arrival of an angry paramour who, like the bike path obstacle, represents a violation of the predicted course that demands rapid braking (and perhaps a switch to self-protective motivations).

Note here that this grounded perspective has taken social and emotional factors seriously (e.g., Niedenthal, Barsalou, Winkielman, Krauth-Gruber, & Ric, 2005) but has shied away from the functional analysis of motivational and emotional systems key to the present account. Thinking in terms of fundamental motivations provides an organizing framework and deepens the grounding while respecting other recent advances.

Dynamically implemented archetypes as strange attractors. The subsymbolic tools we have discussed in this section evoke many of Jung's original formulations (e.g., that they are "inexhaustible" in the specific forms that they take; that they give rise to both images and "transformations"). Subsymbolic implementations do have limitations, but that should not conceal the paradigm shift they have ushered in for our understanding of mental representation, and once they are computationally grounded in perception and action dynamics, other tools of the dynamical systems approach to cognition (e.g., Killeen, 1992; Thelen & Smith, 1996) become inevitable.

One of these is to characterize mental representations as having *attractors*—centers of gravity that pull the perception of otherwise ambiguous stimuli in certain directions rather than others. Simple attractors might show up as potential wells that inevitably draw interpretations to their lowest point, like ball bearings rolling on a landscape. But more often the attractors yield chaotic oscillations that circle a point without ever collapsing to a final state, repeating general trajectories without every exactly repeating a specific trajectory (Abraham & Shaw, 1992). In chaos theory and complexity science, these are called *strange attractors*, and such dynamic trajectories describe the archetypes in ways quite consistent with the three themes we have outlined thus far.

Ontogenetically, Thelen and Smith (1996) showed how early sensorimotor dynamics in the infant reveal strange attractors. In reaching, for example, infants seem to be biologically prepared to first explore the degrees of freedom of the shoulder (with quite erratic movements if we focus on their fingers), then the elbow, then the wrist and fingers. Each phase facilitates a progressive mapping of action onto visual perception. Physically plotting these motions shows orderly reaching dynamics emerging out of chaotic initial behaviors that nevertheless seem to oscillate around the end goal of grasping. It is easy to observe such overt movement dynamics; the archetypes reflect a similar tendency of internal representational systems to gravitate toward certain behaviors of information pick-up in the service of social goals.

Proximate online dynamics can also be illuminated by strange attractors. The subsymbolic model of category learning put forward by Knapp and Anderson (1984) has exactly this character. With small numbers of exemplars experienced repeatedly, the model builds set of attractors that can discriminate between specific instances—for example, the faces of one's family members and close friends. But with many exemplars, each encountered only rarely, a general prototype can also emerge—for example, of the dangerous other, or the ideal mate (which should not be a specific relation, for obvious reasons). With such a model, when we encounter a face, the attractors could pull us to specific identity if the face is familiar, but strange faces will show greater pull toward prototypical modes of activity. It should be noted that even specific identities also have prototypical forms—imagining the face of your grandmother evokes something that is the average of one's experiences, which never exactly repeat themselves—but the broader principle is that novel identities will be more affected by social goal prototypes (e.g., the archetypes). Kelso (1995) wrote extensively about the ways in which concepts can be described as strange attractors. Thinking in terms of strange attractors is thus quite profitable when describing online dynamics and developmental/ontogenetic processes that yield these mature attractor spaces.

Of significance, however, few grounded computational or dynamical systems researchers have seriously considered the motivational and emotional systems that have coevolved with our ability to mentally represent and simulate the important features of the world (but see Killeen, 1992). The social and cultural forces that our minds are designed to expect (and indeed depend on) also rarely make an appearance. It is for this reason that the archetypes provide a useful impetus not only for evolutionary social cognition but also for mainstream cognitive science. They can provoke new theorizing and hypotheses (see next) as we struggle to demarcate their types, their consequences, and the order of their emergence. They also provide a bridge between the dynamical systems approach to proximate, online cognition that is to be found in the grounded simulations and subsymbolic perspectives and the ultimate dynamics of natural selection. Our model of the archetypes maintains that broad architectural properties of the nervous system have coevolved with perception and action regularities to yield a reliably occurring order (a phase space of strange attractors) in the adult mind.

Conclusions

As the increasingly nuanced findings from evolutionary, social, developmental, and cognitive psychology make clear, traditional conceptualizations of representation struggle to account for the sophisticated ways in which people think about the social world. A new conceptualization of social-cognitive representational systems is needed, and we have argued that an enhanced view of Jungian archetypes characterizes important pieces of the emerging solution.

Reviving the Archteypes affords many useful advantages:

1. **It provides a structure within which one can understand a broad range of existing findings.** Neuroscience, for example, now recognizes that the resting brain is almost as active as when it is engaged in cognitively demanding tasks (Raichle & Snyder, 2007).

The significance of this default resting mode (or "daydreaming") remains controversial, but it has a clear (albeit distributed) anatomical structure that includes regions known to be active during social cognition (e.g., the ventromedial prefrontal cortex). A growing number of researchers explicitly link this Default Mode Network activity to social simulations and imagination (e.g., Buckner, Andrews-Hanna, & Schacter, 2008). Indeed, consciousness itself seems to involve a volley of activity between the Default Mode Network (DMN) and various task-positive networks that involve more engaged processing, including attention and executive functioning (Neuner et al., 2014). This is consistent with the perspective advanced here in which domain-specific motives use domain-general resources to achieve their goals. Thus, even in the absence of environmental cues, when we relax and turn inward, the brain is busily running simulations of social scenarios—and very likely those most relevant to recent experiences and anticipated challenges. The Default mode network may reflect the fact that the brain is constantly exercising the archetypes. This is one example, but many other findings fit well with our model.

2. **It integrates conceptual perspectives too often seen as disparate, thereby furthering scientific consilience.** The Archetypes are the natural consequence of fundamental social goals playing out in three nested dynamics, each of which has traditionally been studied by siloed disciplines. The online representation of reality by dynamic, multimodal, and subsymbolically grounded mental simulation systems is the concern of a core subset of cognitivists including those espousing the "embodied" perspective. Few of them ever consider the history of personal experiences that build a particular instantiation of these systems, perhaps because this is seen as the purview of developmental and personality theorists. But the algorithms that operate on these personal histories illuminate the mature forms that our representations take. And then there are the evolutionary dynamics that selected for our cognitive and affective capacities; these are paid some lip service by the cognitive and developmental disciplines, but it has largely fallen to evolutionary psychologists, who themselves rarely tread into the thornier parts of the cognitive thicket. This is despite the fact that much of adaptive behavior seems to be accomplished "on the fly" and so requires a sense of what the proximate mechanisms are doing. Evolutionary perspectives in turn provide a sorely needed "deep structure" for subsymbolically grounded models.[4] Our revival of the archetypes has been driven by an increasing need to draw on these three disparate perspectives. Our model highlights an underappreciated consilience between evolutionary psychologists, developmentalists, and the grounded-computational variants of cognitive science (we are confident that it can ground more abstract symbolic/propositional models as well).

3. **It accounts for findings difficult to explain cleanly otherwise.** Consider our argument that the archetype of aggression is an outgroup male; in one of our studies (Becker et al., 2010), a scary film clip of a White aggressor led White participants to better encode not White male strangers but Black male strangers. A simple associative priming explanation would not predict this, but it is consistent with the idea that self-protection concerns activate an archetype of aggression, energizing encoding resources when stimuli match this template. This archetype can have features that do not match those of the prime but do reflect functional covariation in the participant's personal history, including aspects driven by media and other social fictions. This example speaks to a broader advantage of our approach. There are many kinds of fear, and asking a person to write about a time when they were afraid might make one individual think of terrorists while another ruminates about public speaking. Manipulations that induce fear in a general way will enter these individual differences into the experiment as nuisance variables. "Priming"[5] a fear of interpersonal aggression is more precise and is more likely to lead all participants to evoke the same archetypal representations—likely young men of the outgroup. There may be remaining variance to account for in the specific forms that these take (e.g., militant White supremacist vs. Islamic Jihadist), but other functional commonalities (e.g., male) will show less variance. The Archetypes suggest that ancestrally recurring social problems and opportunities have selected for simulation systems that are highly amenable to such manipulations and lead to nuanced predictions about the specific targets that will be affected.

4. **Finally, it suggests new methodogical emphases that generate novel predictions.** There are many fertile directions in which the Archetypes lead; let us consider three:

    a. Archetypal representational systems are designed to deal with variable environments and still produce forms robust against functionally irrelevant variance. This entails that early experiences can divert the form of archetypes in functional directions. For example, early experiences with resource scarcity or violence will generate different balances of power

[4]One frequently cited limitation of subsymbolic/connectionist models is that they do not "scale up"—they can model simple things like word recognition, but they have yet to give rise to anything like full-blown artificial intelligence. Although it may stymie human programmers, natural selection has had ample time to build structures and strategies that use subsymbolic computation as a design principle at the microscale but motivationally innervated and functionally structured goal-pursuit systems on a larger ecology-tuned scale. Some recognize these potentials (e.g., Thagard, 2006), although it is still a rare position.

[5]Kahneman criticized the "priming" literature, but many of the "failures to replicate" that concerned him use priming methods that are too general (e.g., a crossword puzzle with several words that connote elderly; the warmth or texture of an object). Archetypes live at a sweet spot reflecting specific social challenges, and this makes them eminently primeable. Kahneman was not criticizing all of priming; many associative, semantic, and even affective priming effects are highly replicable. We suggest adopting those kinds of methods (e.g., ideal strategy manipulations; Stone & Van Orden, 1994).

in the affordance management processes that jockey among the archetypes. Methodologically, we should pay attention to individual difference variables that would have plausibly moderated the adult forms of the archetypes. Furthermore, we should discriminate among stimulus-level variables as a function of how reliable they occurred across ancestrally variable environments. For example, our template of threat should reliably include facial expressions of anger and sexually dimorphic cues to masculinity, but be more variable in what constitutes the "outgroup." The Archetypes thus should have certain predictable qualities, regardless of the environment, but other features subject to considerable variability. These can be theoretically derived, those predictions tested, and unsuccessful hypotheses and theories rejected.

b. Consider the order of emergence of archetypal discriminations in the developing infant. This can be assessed in that same way that much of infant research proceeds: by studying where infants look and for how long. Early in development, Archetypes will be all about where to look, and attention at one stage prepares the possibility for new kinds of attention at later stages. Young children may, for example, perseverate on waist-to-hip-ratio (WHR) and shoulder-to-height-ratio (SHR) in women and men, respectively, long before the mating motives come online—this is easily testable and, even if it proves not to be true, will focus future research efforts in other profitable directions.

c. The notion that archetypes are strange attractors also suggests that reaction times can be used as potent indicators of accessibility, as they have been in prototype research (Rosch et al., 1976). Here we may be able to sharpen the distinction between prototypes and archetypes. If researchers contrast culturally arbitrary prototypes of beauty with theoretically universal fitness-relevant features, the accessibility of the latter should be greater despite more potential exposure to the former. Just as measures or retrieval efficiency could reveal the influence of archetypes, short- and long-term memory storage could also reveal that the system is biologically prepared to encode certain stimuli relative to others.

We have tried to weave together a picture of the archetypes that reveals their resonance with some of the most important themes in modern cognitive science. Archetypes indeed generate "images of [social] instincts," as Jung maintained, but archetypes are, more fundamentally, representational systems that afford complex internal simulations. These simulations are grounded in the sensory and motor processes we use to navigate the present world but are mediated and coordinated by more ancient motivational forces. They have an automaticity and a spontaneity that pervades our conscious experience. Each social problem or opportunity rehearses its domain so regularly and routinely that it carves out central tendencies that come to act as attractors, drawing perception, action, and internal representations in the directions that have served the goal well over our personal and ancestral past.

Our enhanced conceptualization of Jungian archetypes does what Marr (1982) asked of all complex theories. It explains the computational roles that the archetypes play (i.e., archetypes are evolved proclivities to simulate and predict adaptive responses to recurring social problems). It suggests developmental algorithms by which these systems assemble themselves (i.e., it postulates that archetypes are merely capacities and depend critically on experience). Finally, it incorporates highly flexible implementations (i.e., archetypal representational systems are dynamic, multimodal, and subsymbolically grounded). We hope it also provokes new thinking to better explain how the mind so effectively represents the complexities and challenges of social life.

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